An Introduction to Measure Theory - Chapter 1

Emil Marinov

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1 Exercise 1.1.1 - Boolean Closure

Show that if $E, F \subset \mathbb{R}^d$ are elementary sets, then the union $E \cup F$, the intersection $E \cap F$, and the set theoretic difference $E \backslash F := \{x \in E : x \notin F\}$, and the symmetric difference $E \Delta F := (E \backslash F) \cup (F \backslash E)$ are also elementary. If $x \in \mathbb{R}^d$, show that the translate $E + x := \{y + x : y \in E\}$ is also an elementary set.

An **elementary set** is that which is a union of a finite amount of boxes. Thus, let $\bigcup_i E_i$ and $\bigcup_j F_j$ be the unions of the finite number of boxes comprising sets E and F, respectively.

$$E \cup F = \bigcup_i E_i \cup \bigcup_j F_j = \bigcup_i \bigcup_j E_i \cup F_j \tag{1}$$

$$E \cap F = \bigcup_i E_i \cap \bigcup_j F_j = \bigcup_i \bigcup_j E_i \cap F_j \tag{2}$$

$$E \backslash F = \bigcup_i E_i \backslash \bigcup_j F_j = \bigcup_i \left(E_i \cap \neg \bigcup_j F_j \right) = \bigcup_i \left(E_i \backslash \bigcup_j F_j \right) \tag{3}$$

This one tripped me up a bit because elementary sets aren't closed under set negation, and I was inherently proving the closure of elementary sets under the set difference for some reason.

$$E \Delta F = E \backslash F \cup F \backslash E = \bigcup_i \left(E_i \backslash \bigcup_k F_k \right) \cup \bigcup_j \left(F_j \backslash \bigcup_k E_k \right) = \bigcup_i \bigcup_j \left(E_i \backslash \bigcup_k F_k \cup F_j \backslash \bigcup_k E_k \right) \tag{4}$$

$$E + x = \bigcup_i (E_i + x) \tag{5}$$

The set theoretical operations of union, intersection, difference, and symmetric difference, and translation, between two elementary sets yield a union of finitely countable boxes, which is itself an elementary set. □

2 Exercise 1.1.2

Give an alternate proof of Lemma 1.1.2(ii) by showing that any two partitions of E into boxes admit a mutual refinement into boxes that arise from taking Cartesian products of elements from finite collections of disjoint intervals.

Lemma 1.1.1(ii) If E is partitioned as the finite union $B_1 \cup ... \cup B_k$ of disjoint boxes, then the quantity $m(E) := |B_1| + ... + |B_k|$ is independent of the partition. In other words, given any other partition $B'_1 \cup ... \cup B'_{k'}$ of E, one has $|B_1| + ... + |B_k| = |B'_1| + ... + |B'_{k'}|$.

$$\begin{aligned} |E| = |\bigcup_k B_k| = |\bigcup_{\tilde{k}} B_{\tilde{k}}| \\ = |\bigcup_k I_{k,1} \times ... \times I_{k,d}| = |\bigcup_{\tilde{k}} I_{\tilde{k},1} \times ... \times I_{\tilde{k},d}| \end{aligned} \tag{6}$$

Let $\{J_{m,1}, ..., J_{m,d}\}$ be the disjoint intervals between all $2k$ endpoints of $\{I_{k,1}, ..., I_{k,d}\}$ and $\{\tilde{J}_{\tilde{m},1}, ..., \tilde{J}_{\tilde{m},d}\}$ be the disjoint intervals between all $2\tilde{k}$ endpoints of $\{I_{\tilde{k},1}, ..., I_{\tilde{k},d}\}$. Let $l_{k,i}$ for $i \in [d]$ denote the disjoint intervals from $J_{m,i}$ that union to form $I_{k,i}$ and $\tilde{l}_{\tilde{k},i}$ denote the disjoint intervals from $\tilde{J}_{\tilde{m},i}$ that union to form $I_{\tilde{k},i}$.

$$|E| = |\bigcup_k \left(\bigcup_{l_{k,1}} J_{l_{k,1}} \times ... \times \bigcup_{l_{k,d}} J_{l_{k,d}} \right)| = |\bigcup_{\tilde{k}} \left(\bigcup_{\tilde{l}_{\tilde{k},1}} J_{\tilde{l}_{\tilde{k},1}} \times ... \times \bigcup_{\tilde{l}_{\tilde{k},d}} J_{\tilde{l}_{\tilde{k},d}} \right)| \tag{7}$$